

Formulas that decide genus

INPUTS

- $|G|$ deck-group order
- n sheet count ($= |G|$ for a Galois cover)
- g_B base-curve genus ($\mathbb{P}^1$, so $g_B = 0$)

1. Projective general linear (PGL) gate

$$\text{pgl}(N) = \max(2N, 60). \quad |G| > \text{pgl}(N) \implies g > 0.$$

Finite subgroups of $\text{PGL} = \text{Aut}(\mathbb{P}^1)$ are bounded by Klein. $|G|$ past that bound forces $g > 0$.

2. Hurwitz automorphism floor

$$\text{For } g \geq 2: |\text{Aut}(C)| \leq 84(g-1) \implies g \geq 1 + |G|/84.$$

Caps the automorphism group; raises the genus floor as $|G|$ grows.

3. Riemann–Hurwitz consistency

$$2g - 2 = n(2g_B - 2) + R. \text{ Over } \mathbb{P}^1: R = 2g - 2 + 2n.$$

Two roles of the formula:

- 1. **Geometric side.** *Given g , it dictates the required total ramification R .*
- 2. **Group-theoretic side (realisability check).** *Does G actually contain a tuple of elements (generating G , multiplying to 1) whose orders yield exactly this R ? If no such assembly exists, the genus g is impossible and must be higher.*

$$|G| \leq \max(2N, 60) \implies g = 0 \qquad \textbf{(PGL)}$$

$$|G| \leq 84(g-1) \implies g \geq 2 \qquad \textbf{(Hurwitz)}$$

$$\text{else} \implies g = 1 \textbf{(small elliptic bound)}$$

Read off g from the bound that $|G|$ satisfies. Riemann–Hurwitz then pins the ramification R .